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To Ban or Not to Ban: Managing the Livestreaming Service With AIGC Technology

Junjie Zu, Can Sun, Xiangyin Kong, and Juzhi Zhang

***Abstract*—Due to breakthroughs in AI-generated content (AIGC) technology, a new form of livestreaming, the so-called virtual livestreaming broadcast by AI agents automatically, has emerged and gained wide adoption by major online platforms. However, other platforms hold opposite attitudes by restricting or forbidding AIGC technology in livestreaming services. In this article, we analytically investigate when the platform should allow or ban the virtual livestreaming service and how the regulation policy affects the seller's operational strategy and consumer surplus by incorporating the operational flexibility and low cost of the virtual livestreaming. We find that allowing the virtual livestreaming can induce the seller to strategically reduce real-person livestreaming effort. Consequently, the platform should ban the virtual livestreaming only when the real-person marginal effort cost falls between two thresholds (contingent on the off-peak-hour market size), but allow it otherwise. We also find that allowing the virtual livestreaming can always yield a triple-win outcome for the consumers, the seller, and the platform, while banning it can either benefit or hurt the consumers and the seller in the monopoly setting. We then generalize our results with two sellers, showing that our results remain qualitatively valid with small or mild degrees of competition. However, as competition intensifies, banning the virtual livestreaming can only harm sellers. By contrast, allowing the virtual livestreaming can lead to a Prisoner's Dilemma, and *interestingly*, a new win–win–win outcome where all sellers fully rely on AI agents to livestream. Finally, we show that our main results can qualitatively hold in multiple settings.**

***Managerial Relevance Statement*—This article provides practical guidance on when platforms should ban or allow AIGC technology in livestreaming. For platform managers, they should tailor AIGC livestreaming policies based on product customization. Specifically, platforms should allow AI agents to handle standardized products efficiently, while for highly customized products, AIGC livestreaming should primarily complement real-person livestreaming during off-peak hours. For products that require extensive consumer interaction and detailed explanations, platforms should restrict AIGC livestreaming to ensure content quality. Furthermore, while platforms, such as TikTok, ban AIGC livestreaming to protect content quality, they must recognize that such prohibitions can severely distort market welfare and harm sellers' profit margins. Instead, policymakers should design flexible regulatory frameworks that permit AIGC as a critical tool to relieve competition. This article also contributes to the following SDGs: SDG 8, SDG 9, and SDG 12.**

***Index Terms*—AI-generated content (AIGC), competition, effort, livestreaming, platform regulation, pricing.**

I. Introduction

LIVESTREAMING selling has become a popular and important business model around the world. The livestreaming market in the U.S. is estimated to be around $18.5 billion in 2023 and is expected to reach $65.1 billion by 2030.[1] The Chinese livestreaming market is more prosperous, with its market size for 2024 approximately 5.8 trillion yuan.[2] Leading online platforms, such as Amazon, Taobao, JD.com, and Douyin (TikTok China), have all established livestreaming channels to facilitate their real-time online transactions. In recent years, due to breakthroughs in AI-generated content (AIGC) technology, including large-language models (LLMs), a new form of livestreaming has emerged and attracted considerable attention in practice, which we call *the virtual livestreaming*. Instead of relying on a real person, the virtual livestreaming is conducted by an AI agent, which is made to have a human appearance and utilize AIGC technology to automatically generate live content to present, explain, and sell products, as well as to interact with consumers. A famous example is the AI agent of Liu Qiangdong, the founder and CEO of JD.com, whose 1-h virtual livestreaming attracted over 20 million viewers and generated over 50 million yuan in sales on 16 April 2024.[3] See Appendix A of the Supplementary Material for more examples of AI agents on various platforms.

Unlike the real-person livestreaming, the virtual livestreaming can operate 24/7, with extremely low costs (typically below one-tenth) and a comparable conversion rate.[4] Due to its high flexibility and low operational cost, the virtual livestreaming

Received 28 January 2026; revised 21 May 2026 and 27 July 2026; accepted 13 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 72471213, Grant 72101243, Grant 72071191, Grant 72471221, Grant 72201262, and Grant 72531010; in part by the Fundamental Research Funds for the Central Universities under Grant WK2040000112; and in part by the University of Science and Technology of China under Grant FSSF-A-260104. Review of this article was arranged by Department Editor X. Sun. *(All authors contributed equally to this work.) (Corresponding author: Can Sun.)*

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This article has supplementary downloadable material available at https://doi.org/10.1109/TEM.2026.3725492, provided by the authors.

Digital Object Identifier 10.1109/TEM.2026.3725492

[1] [Online]. Available: https://www.grandviewresearch.com/horizon/outlook/live-streaming-market/united-states

[2] [Online]. Available: https://www.chinadaily.com.cn/a/202505/13/WS68229ceda310a04af22bee6a.html

[3] [Online]. Available: https://jdcorporateblog.com/jd-com-debuts-ai-digital-representative-of-founder-richard-liu-during-livestream-drawing-20-million-views-in-under-one-hour/

[4] See Table A1 in Appendix A of the Supplementary Material for a comparison between the real-person livestreaming and the virtual livestreaming in a real field test.

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has been widely adopted by major retail platforms in China and welcomed by many small- and medium-sized sellers, who cannot afford or are unqualified to collaborate with influencers or key opinion leaders (KOLs). As a real example, our industry collaborator is a primary technical provider of the virtual livestreaming service for VIP.com, a Top-5 retail platform in China. After successful experimental tests, the platform has promoted the virtual livestreaming to all sellers by offering free subscriptions since 2022. As another example, JD.com is also providing the virtual livestreaming service to vendors on the platform, and over 70 000 vendors have already subscribed to this service.[5] Taobao, the largest retail platform in China, also provides the virtual livestreaming to its sellers and even permits them to purchase such services from third-party providers.[6]

Despite its advantages and increasing popularity, other platforms instead have a negative attitude toward the virtual livestreaming. For instance, Douyin explicitly prohibits all forms of non-in-person livestreaming, and its Vice President Li Liang stated on his social media that "we do not recommend merchants to purchase virtual livestreaming services."[7] Indeed, it is reported that Douyin banned more than 30 000 accounts in the second half of 2024 alone for their use of the virtual livestreaming.[8] Other platforms, such as Kuaishou and WeChat Channels, are also restricting the virtual livestreaming by banning the shopping functions of accounts once an AI agent is detected.[9] These platforms emphasize the use of real-person hosts, highlighting their ability to interact with viewers in real time, respond to questions spontaneously, and create a more authentic and engaging livestreaming experience. The divergent attitudes and policies across platforms raise the following research question: How should a platform manage AIGC technology in the livestreaming service? Specifically, *under what circumstances should a platform ban or allow the virtual livestreaming?*

The platform's policy inevitably has a profound impact on the seller's livestreaming strategy and consumer surplus. On the one hand, if the platform bans the virtual livestreaming to ensure high-quality content, then sellers can only use the real-person livestreaming. However, the high operating costs of the real-person livestreaming drive many sellers (e.g., on Douyin and Kuaishou) to procure virtual livestreaming services from third-party providers to evade the monitoring and regulation of the platform. On the other hand, allowing the virtual livestreaming gives sellers more flexibility in managing their livestreaming accounts. According to our industry collaborator, many of their sellers use different virtual livestreaming strategies: some of them only use AI agents to fill the off-peak hours when the real-person livestreaming is offline, whereas some sellers fully use AI agents to broadcast in their livestreaming rooms. This leads to our second research question: *How does the platform's regulation of the virtual livestreaming affect the seller's livestreaming strategy and consumer surplus?* Furthermore, when the platform allows the virtual livestreaming, all sellers can access such services. For example, VIP.com offers a free virtual livestreaming service to all sellers. Since sellers are facing fierce competition in practice, it is unclear *how competition can affect the platform's policy on the virtual livestreaming and sellers' livestreaming strategies in equilibrium.*

To address the above questions, we build an analytical model in which a seller sells its product via livestreaming and the platform can either ban or allow the virtual livestreaming. To capture the operational flexibility and low cost of the virtual livestreaming, our model incorporates two key features. First, we divide the livestreaming period into two slots, i.e., peak hours and off-peak hours, and the seller always adopts the virtual livestreaming to fill the off-peak-hour market whenever available. This is consistent with the practice that AI agents can broadcast 24/7, while real-person livestreamers can only work for limited periods, typically during peak hours, due to health and legal concerns. Second, we assume that the seller can exert costly effort to enhance the real-person livestreaming quality. By contrast, we normalize the subscription fee of the virtual livestreaming to zero without loss of generality to capture its cost advantage and relatively low livestreaming quality (we relax this assumption in Section VI-B). Then, if the platform bans the virtual livestreaming, then the seller can only use the real-person livestreaming during peak hours. Otherwise, when the virtual livestreaming is allowed, the seller can either combine the real-person livestreaming with the virtual livestreaming to broadcast during peak and off-peak hours, respectively, or use the virtual livestreaming all the time. We use backward induction to solve the platform's optimal regulation policy and the seller's optimal pricing and effort decisions (we instead consider that the seller decides quantity and effort in Section VI-C). Finally, we investigate the impact of competition by considering a duopoly setting in which two sellers sell through livestreaming simultaneously on the platform.

Our main findings are as follows. *First*, we find that when the virtual livestreaming is allowed, its presence in off-peak hours induces the seller to lower their peak-hour livestreaming effort or quality. This result highlights the strategic interactions between the virtual livestreaming and the real-person livestreaming, which is new to the literature. *Consequently*, the equilibrium in the monopoly setting, including the platform's optimal regulation policy on AIGC and the seller's livestreaming strategy, critically depends on the market size in off-peak hours and the seller's marginal effort cost of the real-person livestreaming. Specifically, the platform should allow the virtual livestreaming when the marginal effort cost is below a lower threshold or exceeds a higher threshold (both contingent on the off-peak-hour market size), while it should ban the service when the marginal effort cost falls between the two thresholds. When the virtual livestreaming is allowed, the seller should use the AI agent to broadcast all the time if the marginal effort cost is low. However, when the marginal effort cost is high, the seller should combine

[5][Online]. Available: https://www.yicaiglobal.com/news/jdcom-reveals-over-70000-sellers-use-its-digital-human-livestreaming-hosts

[6][Online]. See multiple virtual livestreaming service providers permitted on Taobao: https://fuwu.taobao.com/ser/applist.htm?viewMarket=fuwuIt&key=&pageSize=24&startPrice=&endPrice=&catMapIds=206476012&sort=time-desc&currentPage=1

[7][Online]. Available: https://finance.sina.com.cn/tech/digi/2025-03-17/doc-inepyfee0494284.shtml?utm_source

[8][Online]. Available: https://finance.sina.com.cn/tech/shenji/2024-12-11/doc-inczamtn7638515.shtml

[9][Online]. Available: https://www.cls.cn/detail/1707754

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the AI agent with the real-person livestreaming to broadcast during off-peak and peak hours, respectively.

The above equilibrium outcome not only echoes our observations in practice, but also offers useful guidance for the platform and seller in managing the livestreaming service with AIGC technology. For instance, we can interpret the marginal effort cost as a measure of product customization; then our results suggest that the platform should ban the virtual livestreaming for medium-customized products, but allow it for low- or high-customized products. For low-customized or standardized products (e.g., books, pencils, and daily necessities, or famous brands), sellers can fully rely on AI agents to broadcast all the time. Indeed, we do observe that some sellers on VIP.com only use the virtual livestreaming to sell daily necessities. However, for high-customized products or professional products (e.g., high-end sports gear, drones, etc.), it requires consumers themselves to have clear preferences or substantial knowledge about the product, making the two livestreaming modes barely different in attracting consumers. For these products, the platform can encourage the seller to use the virtual livestreaming as a supplement to the real-person livestreaming to explore the off-peak-hour market. As for medium-customized products (e.g., fashion accessories or apparel, beauty and cosmetics, and personal care), consumers may heavily rely on the real-person livestreamer to understand products or services. It is better for the platform to forbid the virtual livestreaming. In fact, on content platforms like Douyin, products in categories, including fashion, beauty, and personal care, contribute to nearly half of all livestreaming sales.[10] Similarly, on Kuaishou, apparel and shoes constantly rank as the top category in livestreaming sales over time.[11] In this sense, our results provide a potential explanation for why content platforms are more inclined to ban the virtual livestreaming.

Second, we find that allowing the virtual livestreaming can always yield a *triple-win* outcome for the consumers, the seller, and the platform, while banning the virtual livestreaming can either benefit or hurt the consumers and the seller. In particular, when the marginal effort cost is below some threshold (contingent on the off-peak-hour market size), banning the virtual livestreaming can still benefit all parties, i.e., a win–win–win outcome remains. However, when the marginal effort cost exceeds the threshold, the seller gets worse off while the consumers are still better off from the absence of the virtual livestreaming, i.e., a win–lose–win outcome arises in this case. As the marginal effort cost keeps increasing and exceeds another threshold (also depending on the off-peak-hour market size), the consumers also get worse off from banning AI agents, yielding a lose–lose–win outcome. These different outcomes caution regulators or platforms to carefully examine their policies on restricting or forbidding AIGC technology in livestreaming.

Third, we show that our results in the monopoly setting remain qualitatively valid under small and mild degrees of competition. However, we uncover three new findings on the equilibrium and social welfare when the competition becomes sufficiently intense. First, in this case, a *Prisoner's Dilemma* can arise when the marginal effort cost is high enough. Indeed, both sellers can gain higher profits by simultaneously using AI agents to livestream during both peak and off-peak hours, while both are trapped in an equilibrium where the virtual livestreaming is only used in off-peak hours. Second, we show that when the degree of competition is sufficiently large, banning virtual livestreaming only harms sellers. *Last and interestingly*, when the competition becomes extremely intense, allowing the virtual livestreaming can yield a new win–win–win outcome where both sellers fully rely on AI agents to broadcast for a medium effort cost and a high off-peak-hour market. This new finding underscores that it may help to understand why VIP.com offers free virtual livestreaming services to all sellers. By promoting AI agents, VIP.com can help sellers to enhance their competitiveness in livestreaming selling and benefit consumers with lower price, eventually enjoying a triple-win outcome itself. Our results then demonstrate the value and capability of the virtual livestreaming in dealing with intense competition by highlighting its low cost and operational flexibility.

Finally, we extend our base model to multiple setting, including cases with the spillover effect between livestreaming channels, the nonzero fixed subscription cost, and quantity competition. We show that our main results can qualitatively hold in these settings and generate additional insights. First, when there is a spillover effect from the real-person livestreaming to the virtual livestreaming channel, we find that a stronger spillover effect makes the virtual strategy more attractive even when off-peak traffic is small. Second, when the platform can charge a nonzero fixed subscription cost for the virtual livestreaming, the platform is more likely to offer the virtual livestreaming service, expanding the corresponding equilibrium region for the seller to use the AI agent. Third, we find that the Prisoner's Dilemma observed in the base model disappears in the presence of quantity. This is because, in quantity competition, sellers tend to reduce their quantity to prevent a price collapse as competition intensifies. This quantity reduction lowers the marginal benefit of effort, leading to optimal effort continuously decreasing. Therefore, sellers never exert excessive efforts, and the Prisoner's Dilemma vanishes.

The rest of this article is organized as follows. We review the related literature in Section II. Section III presents the main model, and Section IV analyzes the equilibrium in the monopoly setting. Section V generalizes the results to the duopoly setting to analyze the impact of competition on the equilibrium. We extend the base model to different settings in Section VI. Finally, Section VII concludes this article. Appendix A of the Supplementary Material provides real examples of AI agents and summarizes the notations. All proofs are relegated to Appendix B of the Supplementary Material. The complete analyses of all extensions are provided in Appendices C1–C3 of the Supplementary Material.

II. Literature Review

Our research is related to the burgeoning literature on livestreaming selling. This stream of literature focuses exclusively on the real-person livestreaming and its adoption and subsequent impact on firms' channels and selling strategies.

[10][Online]. Available: https://ecommercechinaagency.com/ecommerce-what-chinese-consumers-are-buying-like-crazy/

[11][Online]. Available: https://xk.newrank.cn/category/market

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Pan et al. [1] examined platforms' transition from the traditional selling channel to livestreaming by considering demand switching, consumer costs and preferences, and streamer capability. Li et al. [2] further distinguished between third-party, self-run, and both types of livestreaming channels, showing that the adoption of livestreaming selling depends on streamer popularity, costs, consumer preferences, cross-channel effects, and product characteristics. Recently, Wu et al. [3] found that e-tailers' livestreaming channel choice depends on network effects and hassle costs. Gu et al. [4] studied a shoppertainment setting where a streamer with limited bandwidth chooses between sharing product information and providing entertainment, finding that more entertainment can boost consumer attendance, affect pricing, and increase streamer profit. Recently, there is a growing number of studies considering the impact of influencers or KOLs. Several studies investigate whether sellers should employ influencers or rely on their own staff for livestreaming [5], [6], [7]. Other scholars, however, examine the operational challenges for sellers when collaborating with influencers or KOLs in livestreaming selling. For example, Qi et al. [8] categorized influencers into regular and top based on bargaining power and studied the contracting problem between sellers and influencers. Lin et al. [9] showed that in price negotiations with KOLs, raising commissions can be more effective than cutting prices, especially when product availability is low. Pu et al. [10] examined how influencers shape consumer buying and tipping in livestreaming, creating a win–win for both influencers and manufacturers, even when small influencers sell without commission. Yang et al. [11] analyzed how cooperation and associated spillover effects with KOLs influence a dual-channel e-commerce supply chain. Different from the above literature, our work considers the so-called the virtual livestreaming, which is enabled by AIGC technology, and investigates its interactions and management together with the real-person livestreaming. We also examine how the platform should ban or offer such a new livestreaming mode and how sellers should manage their operational strategies with or without the virtual livestreaming.

Our work directly contributes to the literature on livestreaming with AI. Thanks to breakthroughs of AIGC and LLMs, the virtual livestreaming is arguably the most successful application of AI in business during recent years. Despite its increasing popularity, there are only a few studies in the literature. Liu et al. [12] empirically showed that enhanced real-time Q&A between virtual-person and consumers can increase sales. With respect to analytical modeling, Niu et al. [13] considered network externalities to explore the tradeoffs between the virtual livestreaming and KOL livestreaming for global brands. Wei et al. [14] indicated that consumers' hassle costs drive the optimal choice between virtual hosts and KOLs. Xu et al. [15] examined the adoption of AI livestreaming in agency and reselling modes and discussed the impact of sales effort, broadcasting power, consumer expectations, and product-fit uncertainty. Zhou et al. [16] indicated that in competition with offline retailers, online retailers' optimal choice between the real-person and the virtual-person livestreaming depends on both the cost of the real-person livestreaming and consumers' hassle costs. Xu et al. [17] examined whether the adoption of AI livestreaming by e-tailers affects supplier encroachment. Our study differs from the above literature in the following aspects. First, we emphasize the operational flexibility of the virtual livestreaming in covering off-peak hours, which is a missing point in the literature but concurred by practitioners we consult. Second, we compare AI agents with average employees instead of influencers or KOLs to focus on the operational strategy of many small- and medium-sized sellers on the platform. Furthermore, we consider the platform's regulation on the virtual livestreaming.

Our work connects to the literature on new technology adoption or regulation in online platforms. The extant literature has examined the impact of these new technologies, such as Internet of Things (IoT), blockchain, and AI, in a variety of settings. Sun and Ji [18] showed that the platform's pricing scheme significantly affects IoT investment, and higher IoT value sometimes reduces profits. In the later work, Sun et al. [19] showed that sharing IoT data can increase profits and reduce encroachment incentives, but does not always benefit the retailer. Choi [20] examined which model is superior between blockchain platform and traditional retail, depending on the tradeoff between blockchain's authentication benefits and retail's shopping convenience in luxury supply chains. Shen et al. [21] showed that blockchain can alter platform preferences for second products and enabled a triple-win outcome for the platform, suppliers, and consumers. Xu et al. [22] showed how blockchain affects supply chain decisions, profits, and coordination with the cross-channel effect. Other scholars consider the impact of AI. For example, Wang et al. [23] showed that AI-powered technology can help e-tailers manage returns and purchase uncertainty, although it may also encourage opportunistic consumer behavior. Cheng et al. [24] explored the optimal AI-based consumer experience tools implementation strategies for the retailer and supplier by considering consumer privacy sensitivity in resale and direct sales channels. Bai et al. [25] revealed that data sharing between platforms and retailers can boost AI service quality and demand but may also create inefficiencies. We contribute to this stream of literature by investigating the adoption and regulation of AIGC technology in livestreaming services on online platforms.

III. Model

A. Livestreaming Mode and Demand

We first consider the monopoly setting where a seller sells the product to consumers on a platform via livestreaming (we shall consider the duopoly setting in the next section). The platform can provide two different livestreaming modes: the real-person livestreaming and the virtual livestreaming, denoted by superscript R and V, respectively. In the real-person livestreaming mode, the seller hires an employee to broadcast. The employee is responsible for illustrating product designs and functions, responding to consumers' queries and guiding consumers to purchase during livestreaming. Hence, the employee's effort, denoted by e, is critical to boosting the demand. For instance, livestreamers in the apparel business usually put on the clothes themselves to show the design. When selling consumer electronics, livestreamers typically elaborate on technical parameters and new functions in detail. A motivated employee often answers

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questions and comments promptly and humorously to entertain consumers and attract more viewers.

The virtual livestreaming, however, is conducted by an AI agent rather than a real person. The AI agent can be trained by collecting photos of real individuals and using AIGC technology to simulate the real-person livestreaming. Compared with the real-person livestreaming, the virtual livestreaming can operate 24/7 with extremely low daily operating cost. It is estimated that the daily operating costs are at least one-tenth or less than that of the real-person livestreaming. However, the virtual livestreaming quality is typically believed to be inferior to that of the real-person livestreaming, due to algorithm aversion (e.g., consumers' resistance to chat with AI agents) or technical limitations (e.g., AI agents may misunderstand viewers' comments). Therefore, sellers in practice are more inclined to adopt the real-person livestreaming during peak hours, e.g., the golden hours from 6 P.M. to 10 P.M. on weekdays, while adopting the virtual livestreaming during off-peak hours with low traffic, such as late nights.

To capture traffic heterogeneity during livestreaming and different broadcasting strategies for sellers, we divide the livestreaming period into two slots: one is *peak hours* with high traffic, and the other is *off-peak hours* with relatively low traffic. We normalize the mass of consumers during peak hours to one, and let $\alpha < 1$ denote the mass of consumers during off-peak hours. In the following, we use the subscripts g and o to denote peak hours and off-peak hours, respectively. We also refer readers to Table A1 in Appendix A of the Supplementary Material for a real example provided by our industry collaborator, which is consistent with the above setup on different time slots.

If the seller adopts the real-person livestreaming, then we assume that the following inverse demand function for peak hours:

$$p = 1 + e - q_g^R \tag{1}$$

where e denotes the employee's effort in the real-person livestreaming. The additive form of effort captures the demand boosting effect, which can also represent the relatively higher quality of the real-person livestreaming than the virtual livestreaming. Note that the above linear demand can be derived by classical quadratic utility [see e.g., (CS), which measures the consumer surplus] and is commonly adopted in the literature, e.g., Singh and Vives [26], Raju et al. [27], Abhishek et al. [28]. We also refer readers to Appendix B of the Supplementary Material for the microfoundation and detailed derivation for the demand functions.

When the virtual livestreaming is available, the seller can subscribe to this service from the platform and broadcast automatically. Similar to the real-person livestreaming, we assume the following inverse demand functions for the virtual livestreaming during peak and off-peak hours, respectively:

$$p = 1 - q_g^V \tag{2}$$

$$p = 1 - q_o^V. \tag{3}$$

That is, the seller faces the same linear demand during peak and off-peak hours when using the virtual livestreaming. This captures the consistent but relatively lower quality of this livestreaming mode.

We next model the problems of the platform and the seller.

B. Strategies of the Platform and the Seller

In practice, different platforms have different attitudes toward the virtual livestreaming. As aforementioned, many retail platforms, such as JD.com, VIP.com, and Taobao, allow the virtual livestreaming, with some even providing a free service (we consider that the platform charges a fixed subscription fee for service in Section VI-B). By contrast, some platforms, such as Douyin and Kuaishou, ban AIGC technology in livestreaming. Hence, platforms in practice need to decide how to regulate AIGC technology in livestreaming. To simplify the analysis, we assume that the platform either allows or bans the virtual livestreaming service.

1) If the platform bans the virtual livestreaming, then the seller can only use the real-person livestreaming. We further assume that the seller never adopts the real-person livestreaming during off-peak hours because of the high effort cost and legal requirements. This assumption is reasonable and consistent with industry practice. Livestreaming is a mentally exhausting activity, so many livestreaming rooms, to ensure broadcast quality, stream only for limited durations, such as 4–6 h. Meanwhile, labor laws typically stipulate that a person's normal daily working hours are 8 h.

In this case, the seller only broadcasts during peak hours and needs to decide the selling price p and livestreaming effort e. We refer to this case as the *RN* strategy. The seller's problem can be formulated as follows:

$$\pi_s^{RN} = \max_{p,e} (1 - \lambda) p q_g^R - k e^2 \tag{4}$$

where λ denotes the commission rate charged by the platform for each transaction. We assume that the commission rate is fixed, which is commonly adopted in the platform operations literature; e.g., Lin et al. [9], Abhishek et al. [28], Rayan et al. [29], and Hagiu and Wright [30]. In practice, many large platforms, e.g., Amazon, Taobao, and Douyin, charge different fixed commission rates for different categories of products.[12] The parameter k denotes the seller's marginal effort cost, which can be interpreted as the difficulty of elaborating the product functions. The quadratic form of the effort cost is also common in the literature (e.g., Li et al. [2], Chen and Xiao [31], and Khanjari et al. [32]), which captures the idea that delivering higher livestreaming quality requires substantial additional effort cost. Note that we abstract away the seller's contract design problem in hiring an employee to focus on the strategic choice of livestreaming modes, rather than on the potential moral hazard arising from the seller's delegation of livestreaming operations.

[12]We give some real examples from major platforms showing that commission fees are predominantly determined by product categories rather than each product, such as Amazon (https://feedvisor.com/university/referral-fee/), Taobao (https://bk.taobao.com/k/taobaozhibo_194/f8e705d9b515a6774e9dbffd7dc9d0c7.html), and Douyin (https://m.maijiaw.com/article/633769).

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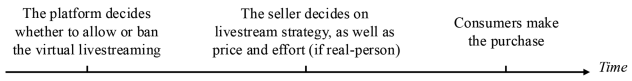


Fig. 1. Game sequence.

Accordingly, the platform's profit in this case is

$$\pi_p^{RN} = \lambda p q_g^R. \tag{5}$$

2) If the platform allows the virtual livestreaming, then the seller can now subscribe to the virtual livestreaming from the platform. The seller can use an AI agent to automatically broadcast during peak and off-peak hours. We normalize the subscription fee to zero without loss of generality. In practice, the subscription fee is a constant and much less than the operating cost of the real-person livestreaming; thus, it can be ignored. We also assume that the seller will adopt the virtual livestreaming during off-peak hours whenever available. It is intuitive and consistent with the practice. Since there is zero cost to broadcast, adopting the virtual livestreaming can at least attract some consumers during off-peak hours. According to our collaborating platform, most sellers shift from the real-person livestreaming to the virtual livestreaming right after peak hours. Some sellers even run it all day long.

The seller has two strategies: one is to use the real-person livestreaming during peak hours and the virtual livestreaming during off-peak hours. We refer to this strategy as the *RV* strategy. The other is to adopt the virtual livestreaming for both peak and off-peak hours, which we refer to as the *VV* strategy. Note that we do not consider the option that the seller sells through the livestreaming channel of influencers or KOLs. This allows us to focus on average users of the virtual livestreaming, such as small- and medium-sized sellers on the platform, and to rule out the advertising and signaling effects of KOL livestreaming.

For the RV strategy, the seller needs to decide the livestreaming effort e for peak hours and the selling price p. Hence, the seller's problem can be formulated as follows:

$$\pi_s^{RV} = \max_{p,e} p(q_g^R + \alpha q_o^V)(1-\lambda) - ke^2 \tag{6}$$

where $q_g^R + \alpha q_o^V$ is the total demand from peak and off-peak hours. Correspondingly, the platform's profit in this case is

$$\pi_p^{RV} = \lambda p(q_g^R + \alpha q_o^V). \tag{7}$$

For the VV strategy, the seller only needs to decide the selling price p, and the problem is as follows:

$$\pi_s^{VV} = \max_{p} p(q_g^V + \alpha q_o^V)(1-\lambda). \tag{8}$$

The platform's profit in this case is

$$\pi_p^{VV} = \lambda p(q_g^V + \alpha q_o^V). \tag{9}$$

C. Game Sequence

As shown in Fig. 1, the game sequence is as follows.

1. The platform decides whether to allow or ban the virtual livestreaming.
2. Following the platform's decision, the seller determines the strategy.
   i) If the platform bans, then the seller can only use the real-person livestreaming in peak hours and decides the selling price p and the livestreaming effort e.
   ii) If the platform allows, then the seller decides how to use the virtual livestreaming in two time slots.
      a) The seller can use the real-person livestreaming in peak hours and the virtual livestreaming in off-peak hours, respectively. The seller decides the selling price p and the livestreaming effort e.
      b) The seller can use the virtual livestreaming both in peak and off-peak hours. The seller decides the selling price p.
3. Consumers make the purchase.

IV. Analysis

In this section, we analyze the equilibrium using backward induction. We first analyze the subgame equilibrium for the seller's optimal selling price and livestreaming effort (if available), given that the platform allows or bans the virtual livestreaming. We then analyze the platform's optimal decision on when to allow or ban the virtual livestreaming.

A. Subgame Equilibrium: The Seller's Optimal Decision

We first analyze the case when the platform bans the virtual livestreaming. In this case, the seller can only broadcast during peak hours using the real-person livestreaming, i.e., the RN strategy, and decide the optimal selling price and livestreaming effort.

There exists a threshold $k_1 = \frac{1-\lambda}{4}$, and we have the following result for the RN strategy.

Lemma 1: If the platform bans the virtual livestreaming, then the seller's problem is jointly concave and the RN strategy is feasible given $k > k_1 = \frac{1-\lambda}{4}$. The seller's optimal decisions are $p^{RN} = \frac{2\tilde{}k}{4k-(1-\lambda)}$ and $e^{RN} = \frac{1-\lambda}{4k-(1-\lambda)}$. The corresponding demand and profit for the seller are $q^{RN} = \frac{2\tilde{}k}{4k-(1-\lambda)}$ and $\pi_s^{RN} = \frac{k(1-\lambda)}{4k-(1-\lambda)}$. The platform's profit is $\pi_p^{RN} = \frac{4\tilde{}k^2\lambda}{(4k-(1-\lambda))^2}$. Furthermore, the optimal price, effort, the corresponding demand, and the seller's profit all decrease in k and λ. The platform's profit decreases in k, while it decreases in λ when $k_1 < k < \frac{1+\lambda}{4}$ but increases in λ when $k > \frac{1+\lambda}{4}$.

Lemma 1 indicates that the real-person livestreaming only becomes feasible when the marginal effort cost k is relatively high ($k > k_1$). Note that, when $k < k_1$, the seller's optimization problem is no long concave, resulting in $e \to \infty$ or $e = 0$. In practice, real-person livestreaming requires efforts (e.g., preparation and time spent) from the real streamer, and such effort is bounded by physical stamina, finite broadcasting capacity, and legal working hours.

Given the RN strategy is feasible, the seller's optimal price and effort for the RN strategy both decrease as the marginal effort cost k increases. Consequently, the seller's profit also decreases in k. This should be intuitive. As k increases, real-person livestreaming becomes more costly and less effective, making the seller reduce the optimal effort and price. The lower

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effort reduces the demand while the lower price can increase the demand. Eventually, the effect of effort dominates the price effect, leading to a lower demand and a lower profit for the seller. It is also straightforward that the seller earns less profit as the platform charges a higher commission rate, i.e., λ increases.

Lemma 1 also indicates that the platform does not necessarily benefit from charging a higher commission rate when the marginal effort cost is relatively small ($k_1 < k < \frac{1+\lambda}{4}$). The logic behind this result is as follows. Recall that the increase in the commission rate drives the seller to reduce price and effort, resulting in a lower demand. When the marginal effort is small, a higher commission rate cannot compensate for the negative effects caused by the decreasing price and demand. Hence, the platform's profit first decreases then increases in λ.

We next turn to the case when the platform does not ban but offers the virtual livestreaming service to the seller. When the virtual livestreaming is available, the seller can either 1) adopt the real-person livestreaming during peak hours then shift to the virtual livestreaming during off-peak hours, i.e., the RV strategy, or 2) use the virtual livestreaming to broadcast all the time, i.e., the VV strategy.

We first present the equilibrium for the RV strategy, as summarized in the following lemma. There exists a threshold $k_2 = \frac{1-\lambda}{2(\alpha+1)} > k_1$, and we have the following result for the RV strategy.

Lemma 2: If the platform offers the virtual livestreaming service, then the seller's problem is jointly concave and the RV strategy is feasible given $k > k_2$. The seller's optimal decisions are $p^{RV} = \frac{2(1+\alpha)k}{4(1+\alpha)k-(1-\lambda)}$ and $e^{RV} = \frac{(1+\alpha)(1-\lambda)}{4(1+\alpha)k-(1-\lambda)}$. The corresponding demand and profit for the seller are $q^{RV} = \frac{2k(1+\alpha)^2}{4(1+\alpha)k-(1-\lambda)}$ and $\pi_s^{RV} = \frac{k(1+\alpha)^2(1-\lambda)}{4(1+\alpha)k-(1-\lambda)}$, and the platform's profit is $\pi_p^{RV} = \frac{4^-k^2(1+\alpha)^3\lambda}{(4(1+\alpha)k-(1-\lambda))^2}$. Furthermore, the optimal price and effort decrease in $k, \lambda,$ and α, and the corresponding demand and seller's profit decrease in k and λ, but increase in α. The platform's profit decreases in k, and decreases in λ if $k_2 < k < \frac{1+\lambda}{4(1+\alpha)}$ and $\frac{1}{3} < \lambda < 1$ but increases otherwise. In addition, the platform's profit decreases in α when $k_2 < k < \frac{3(1-\lambda)}{4(1+\alpha)}$ but increases in α when $k > \frac{3(1-\lambda)}{4(1+\alpha)}$.

Lemma 2 indicates that the feasible threshold k_2 decreases in α but $k_2 > k_1$. Intuitively, a larger off-peak market α generates supplementary demand, making the RV strategy profitable. However, $k_2 > k_1$ indicates that the feasible region for the RV strategy shrinks compared to that of the RN strategy. This is because the RV strategy expands off-peak market demand but forces the seller to lower the price ($p^{RV} < p^{RN}$), which diminishes the marginal benefit during peak hours. Consequently, a higher baseline boundary ($k_2 > k_1$) is required to prevent overinvestment in effort compared to the pure real-person strategy.

Lemma 2 confirms the intuition that the seller's profit indeed increases as the off-peak-hour market size α increases. *By sharp contrast*, Lemma 2 further suggests that the platform can actually benefit for a relatively high marginal effort cost (i.e., $k > \frac{3(1-\lambda)}{4(1+\alpha)}$) as more consumers appear in off-peak hours (i.e., α increases) but *get hurt* for a relatively low marginal effort cost ($k_2 < k < \frac{3(1-\lambda)}{4(1+\alpha)}$).

To explain this counterintuitive result, we first analyze the impact of α on the seller's optimal decisions. First, a larger α implies a larger off-hour demand q_o^{RV}, which positively contributes to the total demand, thus benefiting both the seller and the platform. Second, the seller has an incentive to reduce the selling effort during peak hours when facing a larger off-peak-hour market size α, which negatively affects the peak-hour demand q_g^{RV}. Consequently, considering both effects, the increase in α leads to a rise in off-peak demand q_o^{RV} that outweighs the reduction in peak-hour demand q_g^{RV}, resulting in an overall increase in total demand (i.e., $\frac{\partial q_i}{\partial \alpha} > 0$). Third, as α increases, it is intuitive that the seller would like to reduce the selling price.

Next, we explain why the increase in α leads to divergent profit outcomes for the seller and the platform. For the seller, reducing the selling effort significantly alleviates the effort cost burden. The benefit from the expanded total demand, combined with the substantial cost savings, strictly outweighs the revenue loss from the lower selling price, ensuring that the seller's profit always increases with α. By contrast, the platform's revenue depends on the total sales revenue (i.e., price $\times$ total demand) and does not benefit from the seller's effort cost savings. When $k_2 < k < \frac{3(1-\lambda)}{4(1+\alpha)}$, the seller reduces both effort and price. The negative effect of the lower price dominates the positive effect of the demand expansion, leading to a decrease in total sales revenue, which hurts the platform. Conversely, when the marginal effort cost is relatively high (i.e., $k > \frac{3(1-\lambda)}{4(1+\alpha)}$), the seller's effort is already low, making the subsequent reduction in effort and price marginal. In this case, the demand expansion effect dominates, increasing the total sales revenue and benefiting the platform.

Lemma 2 also indicates that the marginal effort cost k and the commission rate λ have similar impacts on the optimal decisions under the RV strategy as those under the RN strategy. Consequently, the impacts on profits of the seller and the platform are also similar to those for the RN strategy. This should not be surprising since the RN strategy is indeed a special case of the RV strategy when $\alpha = 0$.

We next present the equilibrium for the VV strategy. In this case, the seller uses an AI agent to broadcast during both peak and off-peak hours, hence only needs to decide the selling price.

Lemma 3: If the platform offers the virtual livestreaming service, then the VV strategy is always feasible for the seller. The seller's optimal decision for the VV strategy is $p^{VV} = \frac{1}{2}$ and the corresponding demand is $q^{VV} = \frac{\alpha+1}{2}$. The seller's profit is $\pi_s^{VV} = \frac{(1+\alpha)(1-\lambda)}{4}$ and the platform's profit is $\pi_p^{VV} = \frac{(1+\alpha)\lambda}{4}$. Furthermore, the seller's (respectively, the platform's) profit decreases (respectively, increases) in λ, and the profits of both the seller and the platform increase in α.

Lemma 3 suggests that the seller can always use the virtual livestreaming for continuous broadcasting. Recall that the real-person livestreaming is *not* feasible when the marginal effort cost is relatively small ($k < k_1$ and $k < k_2$, respectively). Lemma 3 thus reveals another advantage of the virtual livestreaming: It can serve as a reliable substitute for the real-person livestreaming during or beyond peak hours, for example, when employees are unqualified or when operating the real-person livestreaming is too costly. Indeed, some sellers on our collaborating platform

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TABLE I
SENSITIVE ANALYSES OF MONOPOLY CASE

	RN			RV			VV		
	$k\uparrow$	$\lambda\uparrow$	$\alpha\uparrow$	$k\uparrow$	$\lambda\uparrow$	$\alpha\uparrow$	$k\uparrow$	$\lambda\uparrow$	$\alpha\uparrow$
p^*	$\downarrow$	$\downarrow$	$\backslash$	$\downarrow$	$\downarrow$	$\downarrow$	$\backslash$	$\backslash$	$\backslash$
e^*	$\downarrow$	$\downarrow$	$\backslash$	$\downarrow$	$\downarrow$	$\downarrow$	$\backslash$	$\backslash$	$\backslash$
q^*	$\downarrow$	$\downarrow$	$\backslash$	$\downarrow$	$\downarrow$	$\uparrow$	$\backslash$	$\backslash$	$\uparrow$
π_s^*	$\downarrow$	$\downarrow$	$\backslash$	$\downarrow$	$\downarrow$	$\uparrow$	$\backslash$	$\downarrow$	$\uparrow$
π_p^*	$\downarrow$	$*$	$\backslash$	$\downarrow$	$*$	$*$	$\backslash$	$\uparrow$	$\uparrow$

Note: $*$ indicates nonmonotonicity, see the corresponding lemmas for details.

have completely shifted from the real-person livestreaming to the virtual livestreaming.

We summarize all sensitive analyses of results in Table I.

Now, we are ready to present the subgame equilibrium by comparing the seller's profits in three strategies.

Proposition 1 (Subgame equilibrium): The subgame equilibrium is as follows.

i) If the platform bans the virtual livestreaming, then the seller chooses the RN strategy with effort e^{RN} and price p^{RN} given $k > k_1$.
ii) If the platform allows the virtual livestreaming, then the seller chooses the RV strategy with effort e^{RV} and price p^{RV} given $k > k_2$; otherwise, the seller chooses the VV strategy with price p^{VV}.

Furthermore, we have $p^{RV} > p^{VV}$ and $q^{VV} < q^{RV}$.

If the virtual livestreaming is banned, then the seller can only use the real-person livestreaming during peak hours, i.e., RN strategy, when marginal effort cost exceeds a certain threshold ($k > k_1$). More importantly, if the virtual livestreaming is allowed, then Proposition 1 suggests that the seller always prefers the RV strategy to the VV strategy whenever the former is feasible (i.e., $k > k_2$, where k_2 critically depends on the off-peak-hour market size α). Indeed, it is common in practice for sellers to use real-person broadcasting during peak hours and then use AI agents to fill off-peak hours. This is due to the demand boosting effect of the real-person livestreaming, which is critical to attract more viewers and potential buyers during peak hours. By broadcasting via a real person, the seller can increase the livestreaming quality during peak hours, allowing the seller to charge a higher price ($p^{RV} > p^{VV}$) and sell to a higher total demand ($q^{RV} > q^{VV}$).

B. Full-Game Equilibrium: When to Ban the Virtual Livestreaming

In this section, we analyze the platform's optimal policy on when to ban or allow the virtual livestreaming. It suffices to compare the platform's profits in both scenarios.

Proposition 2 (Full-game equilibrium): Let $k_3 = \frac{(2+\alpha+\sqrt{\alpha+1})(1-\lambda)}{4(\alpha+1)} > k_2$. Then, the following holds.

i) When $k_1 < k < k_3$, the platform prefers to ban the virtual livestreaming, and the seller chooses the RN strategy.
ii) Otherwise, the platform prefers to allow the virtual livestreaming, and
   a) the seller chooses the RV strategy when $k > k_3$;
   b) the seller chooses the VV strategy when $k < k_1$.

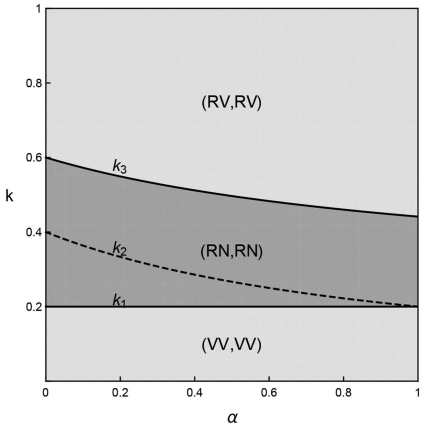


Fig. 2. Platform's decision: banning or allowing the virtual livestreaming ($\lambda = 0.2$).

Furthermore, we have $e^{RN} > e^{RV}$, $p^{RN} > p^{RV} > p^{VV}$, $q^{VV} < q^{RV}$, $q^{RV} >$ (respectively, $<$)q^{RN} if $k > \frac{(2+\alpha)(1-\lambda)}{4(1+\alpha)}$ (respectively, $k_2 < k < \frac{(2+\alpha)(1-\lambda)}{4(1+\alpha)}$), and $q^{RN} >$ (respectively, $<$)q^{VV} if $k_1 < k < \frac{(1+\alpha)(1-\lambda)}{4\alpha}$ (respectively, $q^{RN} < q^{VV}$ if $k > \frac{(1+\alpha)(1-\lambda)}{4\alpha}$). In addition, k_3 decreases in α.

Proposition 2 shows that the platform's optimal policy on the virtual livestreaming critically depends on the marginal effort cost k and the off-peak-hour market size α, as depicted in Fig. 2. When marginal effort cost is extremely small ($k < k_1$), hiring an employee to broadcast during peak hours is not feasible for the seller, while allowing the virtual livestreaming enables the seller to use an AI agent to fill the market in both peak and off-peak hours, which yields a positive profit for the platform. As the marginal effort cost increases ($k_1 < k < k_2$), it is profitable for the platform to ban the virtual livestreaming, i.e., $\pi_p^{RN} > \pi_p^{VV}$. Note that in this case, the RN strategy becomes feasible while the RV strategy is still infeasible, and the seller can only hire an employee to broadcast only during peak hours. Because the marginal effort cost is still relatively low, by focusing on the peak-hour market, the seller can exert sufficient effort to boost demand and charge a higher price. It is more profitable for the platform to ban the virtual livestreaming to enjoy high demand and profit margin in peak hours by giving up the off-peak-hour market. As the marginal effort cost keeps increasing ($k_2 < k < k_3$, where k_3 critically depends on the off-peak-hour market size α), the RV strategy now becomes feasible for the seller and dominates the VV strategy. However, the platform would insist on banning the virtual livestreaming, i.e., $\pi_p^{RN} > \pi_p^{RV}$. This is because, if the seller chooses the RV strategy, he would reduce the selling price and effort due to the presence of the off-peak-hour market. Then, the platform would incur significant demand and profit shortfalls that cannot be compensated through livestreaming during off-peak hours. Finally, when the marginal effort cost is sufficiently high ($k > k_3$), it is more difficult to strengthen the demand through real-person effort. The seller then needs to further reduce the price and effort in peak hours. Facing the thin profit margin, it is crucial for the platform to rely on the additional demand in the off-peak-hour market. Hence,

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the platform prefers to offer the virtual livestreaming service, and the seller chooses the RV strategy.

These results have interesting policy implications for the platform in managing the virtual livestreaming. First, for low-customized or standardized products, e.g., books, pencils, and daily necessities, or famous brands, it is relatively easy for real-person employees to inform consumers of product details (i.e., $k < k_1$). For these products, the platform should allow the virtual livestreaming to replace all the real-person livestreaming. As aforementioned, some sellers in VIP.com fully use AI agents to sell daily necessities all day long. Second, for high-customized or professional products ($k > k_3$), e.g., drones, high-end sports gear, etc., it requires consumers themselves to have clear preferences or substantial knowledge about the product. Hence, the real-person livestreaming is difficult and barely effective in attracting consumers. The platform should encourage the seller to integrate the virtual livestreaming with the real-person livestreaming to explore the off-peak-hour market. Third, for medium-customized products ($k_1 < k < k_3$), e.g., fashion accessories or apparel, beauty and cosmetics, and personal care, consumers may heavily rely on the real-person livestreamer to understand the product or services. It is better for the platform to forbid the virtual livestreaming. Indeed, these types of products prevail in livestreaming sales on content platforms. For instance, on Douyin, fashion, beauty, and personal care products contribute to nearly half of all livestreaming sales, whereas on Kuaishou, apparel and shoes constantly rank as the top category. In this sense, our results offer a potential explanation for why content platforms often ban the AIGC technology in livestreaming.

Another interesting observation from Proposition 2 is that the seller exerts less effort for the RV strategy than the counterpart for the RN strategy ($e^{RV} < e^{RN}$). In other words, the seller may reduce the quality of the real-person livestreaming during peak hours when the virtual livestreaming is allowed. The intuition behind this result is as follows. If the platform allows the virtual livestreaming, then the seller can access the off-peak-hour market via the virtual livestreaming, thus reaching more consumers. As total demand increases, the seller has to lower the price, which reduces both the profit margin and the marginal benefit of effort. Consequently, the seller reduces the real-person livestreaming effort during peak hours. This finding is also consistent with the practice. For instance, according to our industry collaborator, sellers on VIP.com who subscribe to the virtual livestreaming now gradually increase the duration of the livestreaming while shortening the duration of real-person livestreaming even in peak hours. Furthermore, the lower livestreaming effort in the RV strategy leads to a lower peak-hour demand as well as a lower total demand compared with the RN strategy, i.e., $q^{RV} < q^{RN}$, when the marginal effort cost is relatively small ($k_2 < k < \frac{(2+\alpha)(1-\lambda)}{4(1+\alpha)}$). However, for a relatively high $k > \frac{(2+\alpha)(1-\lambda)}{4(1+\alpha)}$, this effect is dominated by the demand increase caused by the lower price, leading to a higher total demand for the RV strategy ($q^{RV} > q^{RN}$). Finally, it is straightforward to see that the seller keeps reducing the livestreaming effort as the marginal effort cost k increases. Consequently, the demand in the RN strategy falls below that of the RV strategy as k increases.

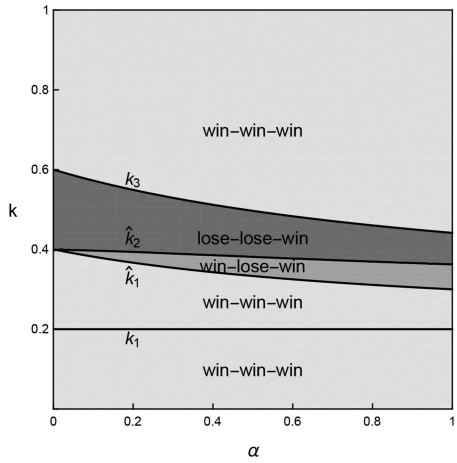


Fig. 3. Welfare analysis of the consumers–sellers–platform ($\lambda = 0.2$).

In the rest of this section, we discuss the impact of the platform's regulation policy on seller's and consumers' welfare. Following Singh and Vives [26], the consumer surplus for the RN strategy given the price p^{RN} and effort e^{RN} can be calculated as

$$\mathrm{CS}^{RN} = (1 + e^{RN})q^{RN} - \frac{(q^{RN})^2}{2} - p^{RN}q^{RN}. \tag{10}$$

Similarly, we calculate the consumer surplus for the RV and VV strategies.

Let $\hat{k}_1$ and $\hat{k}_2$ be defined in Appendix B of the Supplementary Material. We obtain the following results.

Proposition 3: There exists thresholds $\hat{k}_1, \hat{k}_2 \in (k_1, k_3)$ and $\hat{k}_1 < \hat{k}_2$, which is contingent on α, such that:

i) when $k > k_3$ or $k < k_1$, offering the virtual livestreaming service can yield a win–win–win outcome for the consumers, the seller, and the platform;
ii) when $k_1 < k < \hat{k}_1$, banning the virtual livestreaming service can yield a win–win–win outcome for the consumers, the seller, and the platform;
iii) when $\hat{k}_1 < k < \hat{k}_2$, banning the virtual livestreaming can yield a win–lose–win outcome for the consumers, the seller, and the platform; and
iv) when $\hat{k}_2 < k < k_3$, banning the virtual livestreaming can yield a lose–lose–win outcome for the consumers, the seller, and the platform.

Proposition 3 suggests that offering the virtual livestreaming service can always yield a *triple-win outcome* for all parties, as shown in Fig. 3 for $k < k_1$ and $k > k_3$. However, the underlying logic is slightly different for cases when the marginal effort cost is either relatively small ($k < k_1$) or sufficiently high ($k > k_3$). On the one hand, recall that the real-person livestreaming is not feasible when $k < k_1$. Then, all parties can benefit from the virtual livestreaming for fulfilling the vacancy in the market. On the other hand, when $k > k_3$, it is too costly to enhance the peak-hour demand through exerting effort, although the real-person livestreaming (both the RN and RV strategies) is feasible in this case. Hence, the selling effort has less impact, and the profits from the off-peak-hour market become more critical for

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both the seller and the platform. Since the seller chooses the RV strategy when the virtual livestreaming is allowed, the consumer surplus is higher because of the lower price and larger market size, compared with the case when the virtual livestreaming is banned.

Interestingly, Proposition 3 also indicates that banning the virtual livestreaming can have various implications for consumers and the seller, as shown in Fig. 3 for $k_1 < k < k_3$. First, when k is relatively small ($k_1 < k < \hat{k}_1$), banning the virtual livestreaming can also benefit all parties. This is because banning the virtual livestreaming allows the seller to focus on the peak-hour market and exert sufficient effort to enhance demand. Consumers and the seller benefit from the strong demand and high livestreaming quality in peak hours despite the absence of the off-peak-hour market. However, consumers and the seller can *get hurt* by the platform's decision to ban or allow the virtual livestreaming. Specifically, for a relatively high $k(\hat{k}_1 < k < \hat{k}_2)$, the seller gets worse off while the consumers are still better off from the absence of the virtual livestreaming. In this case, the RV strategy becomes feasible, and the seller prefers the RV strategy to the RN strategy (i.e., $\pi_s^{RV} > \pi_s^{RN}$). However, the platform prefers to ban the virtual livestreaming, which hurts the seller. Consumers benefit from banning the virtual livestreaming because they can enjoy a higher livestreaming effort in the RN strategy (i.e., $\mathrm{CS}^{RN} > \mathrm{CS}^{RV}$). Furthermore, as the marginal effort cost keeps increasing ($\hat{k}_2 < k < k_3$), consumers also get worse off from the platform's banning the virtual livestreaming since the seller keeps reducing the selling effort in peak hours.

V. With Competition: Two Sellers

In this section, we investigate how the platform should manage the virtual livestreaming service in the presence of competition. In practice, when the virtual livestreaming is available, many competing sellers subscribe to the service and broadcast during peak or off-peak hours.

To capture the impact of competition, we consider two sellers selling substitute products on the platform via livestreaming. The platform can either offer or ban the virtual livestreaming service. If the platform bans the virtual livestreaming, then seller $i = 1, 2$ can only broadcast via the real-person livestreaming. Seller i then needs to decide the optimal price p_i and selling effort e_i with effort cost ke_i^2. The inverse demand function for seller i using the real-person livestreaming is

$$p_i = 1 + e_i - q_i - bq_{3-i}, i = 1, 2 \tag{11}$$

where $b \in (0, 1)$ denotes the degree of competition, and q_{3-i} is the opponent's demand. If the platform offers the virtual livestreaming service, then seller i can subscribe to the service and only needs to decide the selling price p_i, facing the following inverse demand function

$$p_i = 1 - q_i - bq_{3-i}, i = 1, 2. \tag{12}$$

Clearly, the demand functions in the monopoly setting are special cases of the above when $b = 0$.

Same as in the monopoly setting, there are two time slots: peak hours with a mass of consumers normalized to 1, and off-peak hours with market size $\alpha < 1$. We also assume that each seller always adopts the virtual livestreaming during off-peak hours whenever available and never uses the real-person livestreaming during off-peak hours due to high operational costs or legal regulations. If the platform bans the virtual livestreaming, then each seller can only broadcast using the real-person livestreaming during peak hours, i.e., both sellers choose the RN strategy. Each seller then decides the selling price and effort simultaneously. If the platform allows the virtual livestreaming, then there are three possible scenarios depending on each seller's strategy, as shown in Table II.

TABLE II
Game Matrix for Sellers When the Virtual Livestreaming is Allowed

	RV	VV
RV	$(\pi_1^{RV,RV}, \pi_2^{RV,RV})$	$(\pi_1^{RV,VV}, \pi_2^{RV,VV})$
VV	$(\pi_1^{VV,RV}, \pi_2^{VV,RV})$	$(\pi_1^{VV,VV}, \pi_2^{VV,VV})$

1) Both sellers choose the RV strategy, i.e. broadcast via the real-person livestreaming during peak hours then shift to the virtual livestreaming during off-peak hours.
2) Both sellers use the virtual livestreaming to broadcast all the time, i.e., the VV strategy.
3) One seller, say seller 1, chooses the RV strategy and the other chooses the VV strategy.

Finally, the platform charges a commission fee with a rate $\lambda \in (0, 1)$ for each transaction.

Note that when both sellers choose the same livestreaming strategy, i.e., (RV,RV) and (VV,VV) cases, they face the same demand model. This symmetric demand assumption is widely adopted in the literature, e.g., Xu et al. [17], Singh and Vives [26], and Raju et al. [27]. It allows us to isolate the strategic impact of choosing different livestreaming strategies, instead of the heterogeneity across sellers. This assumption is also consistent with the practice. In VIP.com and Douyin.com, we do observe that many sellers are concurrently selling products from the same supplier or products under the same sales links.

Below, we analyze the game backward. We first investigate sellers' equilibrium strategies given the platform allows or bans the virtual livestreaming. Then, we analyze the full-game equilibrium, focusing on the platform's optimal decision to offer or ban the virtual livestreaming service.

A. Subgame Equilibrium With Competition

Similar to the monopoly case, we first analyze the case when the platform bans the virtual livestreaming. In this case, both sellers broadcast exclusively via the real-person livestreaming during peak hours, denoted by the superscript RN, RN.

In this case, both sellers need to decide the optimal price and selling effort simultaneously, given that the platform bans the virtual livestreaming. There exists a threshold $k_4 = \frac{1-\lambda}{4(1-b^2)}$, and we have the following result for the (RN,RN) strategy.

Lemma 4: If the platform bans the virtual livestreaming, then the RN strategy is feasible for each seller given $k >$

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k_4. For seller $i = 1, 2$, the optimal decisions are $p_i^{RN,RN} = \frac{2(1-b^2)k}{2(2-b)(b+1)k-(1-\lambda)}$ and $e_i^{RN,RN} = \frac{1-\lambda}{2(2-b)(b+1)k-(1-\lambda)}$. The corresponding demand and profit for seller i are $q_i^{RN,RN} = \frac{2\tilde{}k}{2(2-b)(b+1)k-(1-\lambda)}$ and $\pi_i^{RN,RN} = \frac{k(1-\lambda)(4(1-b^2)k-(1-\lambda))}{(2(2-b)(b+1)k-(1-\lambda))^2}$, respectively. The profit for the platform is $\pi_p^{RN,RN} = \frac{8(1-b^2)k^2\lambda}{(2(2-b)(b+1)k-(1-\lambda))^2}$. Furthermore, the optimal effort and the corresponding demand first decrease then increase in b, while the optimal price and the profits of sellers and the platform decrease in b.

Lemma 4 suggests that the degree of competition has a nonmonotonic impact on the optimal effort and, consequently, a nonmonotonic impact on the demand. The underlying driving forces behind the selling effort are twofold. On the one hand, it is straightforward that each seller keeps reducing the price $p_i^{RN,RN}$ as the competition becomes intense (i.e., b increases). This price reduction lowers the marginal benefit of effort, making each seller reduce the selling effort. On the other hand, as the competition becomes intense, it is more critical for each seller to boost the demand by increasing the selling effort. When the degree of competition is relatively small (i.e., $b < \frac{1}{2}$), the first effect dominates; hence, the optimal effort decreases in b. However, for a relatively high degree of competition (i.e., $b > \frac{1}{2}$), the latter effect is stronger, leading to the increase of effort in b. As for the impact on the demand, there are three effects behind. Note that $q^{RN,RN} = \frac{1+e^{RN,RN}-p^{RN,RN}}{1+b}$, then as b increases: first, the substitution effect directly reduces the demand; second, the price reduction increases the demand; and third, the effort effect first reduces then increases the demand. For a relatively small b, the substitution and the effort effects dominate the price effect, leading to a demand shortcut for each seller. For a relatively large b, the price effect and the effort effect dominate then each seller's demand increases.

We next turn to the case when the platform offers the virtual livestreaming service to sellers. We first consider the symmetric cases where both sellers choose the same livestreaming strategy: 1) both sellers choose the RV strategy simultaneously, i.e., both sellers use the real-person livestreaming during peak hours then shift to the virtual livestreaming during off-peak hours; and 2) both sellers choose the VV strategy simultaneously, i.e., both sellers broadcast using the virtual livestreaming during both peak and off-peak hours. Finally, we analyze the asymmetric case where one seller chooses the RV strategy and the other chooses the VV strategy.

The following lemma presents the equilibrium outcome when both sellers choose the RV strategy. In this case, each seller needs to decide the selling price and effort simultaneously. We denote this case by the superscript RV, RV. There exists a threshold $k_5 = \max\{\frac{1-\lambda}{2(\alpha+1)(b+1)}, \frac{1-\lambda}{4(1+\alpha)(1-b^2)}\}$, and we have the following result for the (RV,RV) strategy.

Lemma 5: If the platform offers the virtual livestreaming service, then the RV strategy is feasible for each seller given $k > k_5$. For seller $i = 1, 2$, the corresponding optimal decisions are $p_i^{RV,RV} = \frac{2(\alpha+1)(1-b^2)k}{2(\alpha+1)(2-b)(b+1)k-(1-\lambda)}$ and $e_i^{RV,RV} = \frac{(\alpha+1)(1-\lambda)}{2(\alpha+1)(2-b)(b+1)k-(1-\lambda)}$. The corresponding demand and profit of sellers are $q_i^{RV,RV} = \frac{2(\alpha+1)^2\tilde{}k}{2(\alpha+1)(2-b)(b+1)k-(1-\lambda)}$ and $\pi_i^{RV,RV} = \frac{(\alpha+1)^2\tilde{}k(1-\lambda)(4(\alpha+1)(1-b^2)k-(1-\lambda))}{(2(\alpha+1)(2-b)(b+1)k-(1-\lambda))^2}$, respectively. The platform's profit is $\pi_p^{RV,RV} = \frac{8(\alpha+1)^3(1-b^2)k^2\lambda}{(2(\alpha+1)(2-b)(b+1)k-(1-\lambda))^2}$. Furthermore, the optimal effort and the corresponding demand first decrease then increase in b, while the optimal price and the profits of sellers and the platform decrease in b.

Given that the platform offers the virtual livestreaming, Lemma 5 suggests that the impact of competition when both sellers choose the RV strategy is similar to those in Lemma 4 when the platform bans the virtual livestreaming. It is worth noting that the total demand in the (RV,RV) case also first decreases then increases in b.

We then analyze the other symmetric case where both sellers choose the VV strategy, denoted by the superscript VV, VV. In this case, each seller only needs to decide the optimal selling price.

Lemma 6: If the platform offers the virtual livestreaming service, then the VV strategy is always feasible for seller $i = 1, 2$. The optimal selling price is $p_i^{VV,VV} = \frac{1-b}{2-b}$ and the corresponding demand is $q_i^{VV,VV} = \frac{1+\alpha}{(2-b)(1+b)}$. The profits of sellers and the platform are $\pi_i^{VV,VV} = \frac{(\alpha+1)(1-b)(1-\lambda)}{(2-b)^2(1+b)}$ and $\pi_p^{VV,VV} = \frac{2(1+\alpha)(1-b)\lambda}{(2-b)^2(1+b)}$, respectively. Furthermore, the corresponding demand first decreases then increases in b, while the optimal price and the profits of sellers and the platform decrease in b.

Lemma 6 indicates that the virtual livestreaming can always be a profitable alternative livestreaming choice for each seller to expand the market. Furthermore, competition between sellers can have a negative impact on the platform's profit.

We next analyze the case where the two sellers choose different livestreaming strategies, i.e., one seller chooses the RV strategy and the other chooses the VV strategy. The following lemma presents the equilibrium outcome in this case, denoted by the superscript RV, VV. There exists a threshold $k_6 = \max\{\frac{1-\lambda}{2(1+\alpha)(1-b^2)}, \frac{(\alpha b^2+2\alpha b+b+1)(1-\lambda)}{2(1+\alpha)(1-b)(b+1)(b+2)}\} > k_5$, and we have the following result for the (RV,VV) strategy.

Lemma 7: If the platform offers the virtual livestreaming service, then sellers choose the asymmetric livestreaming strategy given $k > k_6$. The sellers' optimal decisions are given in Appendix B of the Supplementary Material. Furthermore, the optimal effort first decreases then increases in b, while the optimal price and the profits of sellers and the platform decrease in b.

Lemma 7 gives the optimal decisions when sellers adopt the asymmetric strategy. Note that this strategy requires a more stringent feasible condition, i.e., $k_6 > k_5$. It is because when an opponent chooses to use the virtual livestreaming all the time, it can set an extremely low price. The other seller then finds it more difficult and less profitable to match the price while retaining high-quality real-person livestreaming in peak hours. Hence, the (RV,VV) strategy remains feasible only when the marginal effort cost is high enough, i.e., $k > k_6$.

To focus on the effect of competition, we summarize the sensitivity analyses with respect to b in Table III.

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TABLE III
SENSITIVE ANALYSES OF DUOPOLY CASE

	(RN,RN) $b\uparrow$	(RV,RV) $b\uparrow$	(VV,VV) $b\uparrow$
p^*	$\downarrow$	$\downarrow$	$\downarrow$
e^*	$\downarrow\uparrow$	$\downarrow\uparrow$	$\backslash$
q^*	$\downarrow\uparrow$	$\downarrow\uparrow$	$\downarrow\uparrow$
π_s^*	$\downarrow$	$\downarrow$	$\downarrow$
π_p^*	$\downarrow$	$\downarrow$	$\downarrow$

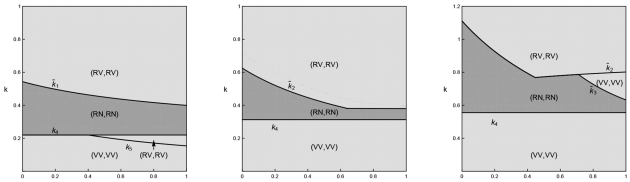


Fig. 4. Platform's decision in duopoly case ($\lambda = 0.2$). (a) $b = 0.3$. (b) $b = 0.65$. (c) $b = 0.8$.

Now, we are ready to present the subgame equilibrium by comparing the sellers' profits. Define $b_1 = 1 - \frac{\sqrt{3}}{3}$, $b_2 = \frac{2}{3}$, and $\hat{k}_3 = \frac{(1-b)(1-\lambda)}{(\alpha+1)(3b^3-5b^2-4b+4)}$, then the results are as summarized in the following proposition.

Proposition 4 (Subgame equilibrium with competition): We have the following results on subgame equilibrium.

i) If the platform bans the virtual livestreaming service, both sellers choose the (RN, RN) strategy only if $k > k_4$.

ii) If the platform offers the virtual livestreaming service, only the symmetric strategies can be sustained in equilibrium.
   a) In equilibrium, both sellers choose the RV strategy if one of the following conditions holds: (1) $k > k_5$ and $0 < b < b_1$; (2) $k > \min\{\hat{k}_3, k_6\}$ and $b_1 < b < b_2$; or (3) $k > k_6$ and $b_2 < b < 1$.
   b) Otherwise, both sellers prefer the VV strategy.
   c) In particular, when $k > k_6$ and $2 - \sqrt{2} < b < 1$, a Prisoner's Dilemma exists, i.e., both sellers choose the RV strategy in equilibrium, while $\pi_s^{VV,VV} > \pi_s^{RV,RV}$.

Furthermore, $p^{RV,RV} > p^{VV,VV}$ and $q^{RV,RV} > q^{VV,VV}$.

There are two interesting observations from Proposition 4. First, the asymmetric (RV,VV) strategy cannot be sustained as an equilibrium. To understand this result, suppose that seller 1 chooses the RV strategy. As we show in the proof, the RV strategy is also feasible for seller 2, and he indeed has incentives to deviate from the VV strategy to the RV strategy (i.e., $\pi_2^{RV,RV} > \pi_2^{RV,VV}$). Similar logic applies if seller 2 chooses the RV strategy and seller 1 has incentives to deviate. Therefore, only symmetric equilibria survive. In other words, it is never optimal for a seller to always use the virtual livestreaming if the opponent uses the real-person livestreaming, and *vice versa*.

Given that only symmetric strategies are optimal, Proposition 4 further suggests that the (RV,RV) strategy becomes the equilibrium only when the marginal effort cost k exceeds some thresholds, which critically depend on the off-peak-hour market size α and the degree of competition b. Otherwise, each seller prefers the VV strategy. This result generalizes Proposition 1 in the monopoly setting. In particular, a Prisoner's Dilemma can arise when the competition is sufficiently intense (i.e., $2 - \sqrt{2} < b < 1$). In this case, both sellers are trapped in the RV strategy even if using the VV strategy yields a higher payoff ($\pi_s^{VV,VV} > \pi_s^{RV,RV}$).

B. Full-Game Equilibrium With Competition

In this section, we compare the platform's profits when it allows or bans the virtual livestreaming service to obtain the full-game equilibrium. The definitions of $\tilde{k}_1, \tilde{k}_2, \tilde{k}_3$, and b_3 are provided in Appendix B of the Supplementary Material, where $b_3 > b_2$.

Proposition 5: We have the following results on full-game equilibrium.

i) The platform prefers to ban the virtual livestreaming, and both sellers choose the RN strategy if one of the following conditions hold: (1) $k_4 < k < \tilde{k}_1$ and $0 < b < b_1$, (2) $k_4 < k < \tilde{k}_2$ and $b_1 < b < b_2$, (3) $k_4 < k < \tilde{k}_3$ and $b_2 < b < 1$.

ii) Otherwise, the platform prefers to offer the virtual livestreaming service, and
   a) if $k > \tilde{k}_1$, $k > \tilde{k}_2$, or $k > \tilde{k}_3$, and especially for the case where $k_5 < k < k_4$ and $b_1 < b < b_2$, both sellers choose the RV strategy;
   b) if $k < \max\{k_4, k_5\}$, and especially for the case where $b > b_3$ and $\tilde{k}_3 < k < \tilde{k}_2$, both sellers adopt the VV strategy.

   Furthermore, we have $p^{RN,RN} > p^{RV,RV} > p^{VV,VV}$, $e^{RN,RN} > e^{RV,RV}$, $q^{RN,RN} >$ (respectively, $<$) $q^{RV,RV}$ if $\max\{k_4, \frac{1-\lambda}{2(1+b)(1+\alpha)}\} < k < \frac{(\alpha+2)(\lambda-1)}{2(\alpha+1)(b-2)(b+1)}$ (respectively, otherwise), and $q^{RN,RN} >$ (respectively, $<$) $q^{VV,VV}$ if $k_4 < k < \frac{(\alpha+1)(\lambda-1)}{2\alpha(b-2)(b+1)}$ (respectively, otherwise).

Proposition 5 suggests that the platform's optimal decision critically depends on the marginal effort cost k, off-peak-hour market size α, and the degree of competition b. To better understand Proposition 5, we show the full-game equilibrium in Fig. 4(a)–(c), where the degree of competition is relatively small ($b = 0.3$), medium ($b = 0.65$), and relatively large ($b = 0.8$), respectively. In each case, as shown in Proposition 5, the platform prefers to ban the virtual livestreaming when the marginal effort cost k is medium (e.g., $k_4 < k < \tilde{k}_1$ when $0 < b < b_1$), but prefers to offer the virtual livestreaming service when k is either low or high (e.g., $k < k_4$ or $k > \tilde{k}_1$, when $0 < b < b_1$). Moreover, the use of the virtual livestreaming leads sellers to charge a lower optimal price and exert less effort (i.e., $p^{RN,RN} > p^{RV,RV} > p^{VV,VV}$ and $e^{RN,RN} > e^{RV,RV}$). These results are in line with those in the monopoly setting and the underlying logic is also similar. Interestingly, Fig. 4 also shows that as the degree of competition b increases, the area where the platform bans the virtual livestreaming moves upward. That is, the platform is more willing to offer the virtual livestreaming service when the marginal effort cost k is small as competition intensifies. This is because the marginal benefit of the real-person livestreaming keeps decreasing as competition

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becomes more intense, rendering the RN and RV strategies infeasible for sellers.

Although the platform's decision appears qualitatively the same with and without competition, Proposition 5 further suggests that there are some subtle contrasts in the sellers' equilibrium strategies in the presence of competition. First, Proposition 5(iia) indicates that both sellers may adopt the RV strategy for a relatively low marginal effort cost, i.e., the (RV,RV) strategy can be sustained as an equilibrium when the degree of competition is low $b < b_1$, as shown in Fig. 4(a) for $k_5 < k < k_4$. Second, Proposition 5(iib) suggests that both sellers may choose the VV strategy for a medium but relatively high marginal effort cost and a high off-peak-hour market α, i.e., the (VV,VV) strategy can be sustained in equilibrium when the degree of competition $b > b_3$, as shown in Fig. 4(c) for $\tilde{k}_3 < k < \tilde{k}_2$. These two equilibria are new due to the presence of competition. Recall that the seller only chooses the VV strategy for a low k and chooses the RV strategy for a high k in the monopoly setting. We next illustrate the rationale behind these two new equilibria.

In the first new equilibrium, the (RV,RV) strategy becomes the equilibrium when k is low and α is relatively large. The reason is that the (RV,RV) strategy can be feasible, i.e., $k_5 < k_4$, when the degree of competition is low ($0 < b < b_1$). Recall that the (RV, RV) strategy always yields higher profit for sellers than the (VV,VV) strategy, both the platform and sellers would prefer to adopt the virtual livestreaming to explore the large off-peak-hour market (α is large). As for the second new equilibrium, competition affects prices for the real-person and virtual livestreaming differently. Note that the real-person livestreaming price is more sensitive to b, i.e., $\frac{\partial p^{RN,RN}}{\partial b} < \frac{\partial p^{VV,VV}}{\partial b} < 0$, hindering the profitability of the RN strategy for each seller. Further note that the demand for real-person livestreaming also becomes limited for a high marginal effort cost k, i.e., $q^{VV,VV} > q^{RN,RN}$ provided $k > \frac{(\alpha+1)(\lambda-1)}{2\alpha(b-2)(b+1)}$ in Proposition 5. Hence, the (VV, VV) strategy yields higher profits for each seller as well as for the platform, $\pi_p^{VV,VV} > \pi_p^{RN,RN}$.

Finally, we investigate the impacts of the platform's decisions on sellers' and consumers' welfare with competition.

Proposition 6: As the degree of competition b increases, the platform's decisions have the following impacts on consumers and sellers.

i) For $b < b_2$, consumer surplus and sellers' welfare resemble those in the monopoly setting.
ii) For $b_2 < b < b_3$, banning the virtual livestreaming yields a win–lose–win outcome for the consumers, the sellers, and the platform; offering it yields a win–win–win outcome for the consumers, the sellers, and the platform.
iii) For $b_3 < b < 1$, banning the virtual livestreaming also yields a win–lose–win outcome for the consumers, the sellers, and the platform; offering it yields a win–win–win outcome for the consumers, the sellers, and the platform.

We show the results of Proposition 6 in Fig. 5(a)–(c), where the degree of competition is relatively low, medium, and high, respectively. First, it is straightforward to see that our findings on consumer surplus and the seller's welfare in the monopoly setting (see Proposition 3) qualitatively hold in the duopoly setting when the degree of competition is relatively low (i.e., $0 < b < b_2$). That is, allowing the virtual livestreaming always benefits consumers and sellers, whereas banning the virtual livestreaming can harm or benefit them, as shown in Fig. 5(a).

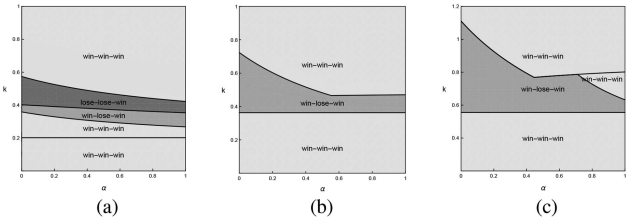


Fig. 5. Welfare analysis of the consumers–sellers–platform in duopoly case ($\lambda = 0.2$). (a) $b = 0.1$. (b) $b = 0.67$. (c) $b = 0.8$.

Fig. 5(b) suggests that as the degree of competition increases (i.e., $b > b_2$), banning the virtual livestreaming only leads to a win–lose–win outcome where only sellers always *get hurt*. Recall that the seller and consumers can get either better or worse off in the low-competition case in Fig. 5(a), sellers can only get worse off due to intensified competition. In this case, the equilibrium prices are highly sensitive to competition (i.e., $\frac{\partial p^{RN,RN}}{\partial b} < \frac{\partial p^{VV,VV}}{\partial b} < 0$), making the (RN,RN) strategy less profitable than the (VV,VV) strategy for sellers. Hence, sellers are worse off if the platform bans the virtual livestreaming. It should not be surprising that consumers can enjoy a cheaper price and thus are always better off from the intensified competition between sellers.

Interestingly, when competition becomes extremely intense (i.e., $b > b_3$), the new (VV,VV) equilibrium in Proposition 5 where the platform allows the virtual livestreaming and both sellers fully rely on AI agents to livestream can also yield a triple-win outcome for all parties when the marginal effort cost is medium and the off-peak-hour market is high enough, as shown in Fig. 5(c). This is in sharp contrast to the monopoly setting where the (VV,VV) strategy can only yield a triple-win outcome when the marginal effort cost k is relatively low.

This new finding highlights the significance and capability of the virtual livestreaming in dealing with fierce competition for sellers and the platform. This finding may help to understand why VIP.com now offers free virtual livestreaming services to all sellers on the platform. It can help sellers to enhance their competitiveness in online selling and benefit consumers with lower prices, and eventually benefit VIP.com itself. In Fig. A2 of Appendix A of the Supplementary Material, we present a real example where the seller livestreams on both VIP.com and Douyin to sell cosmetics (corresponding to a medium but relatively high k). The seller uses the virtual livestreaming and the real-person livestreaming on VIP.com and Douyin, respectively. The price of the virtual livestreaming on VIP.com is 34 yuan, lower than that of the real-person livestreaming on Douyin, 44 yuan. Hence, it is a triple-win outcome for VIP.com to allow the virtual livestreaming.

VI. Extensions

In this section, we consider three extensions to check the robustness of the results and generate additional insights.

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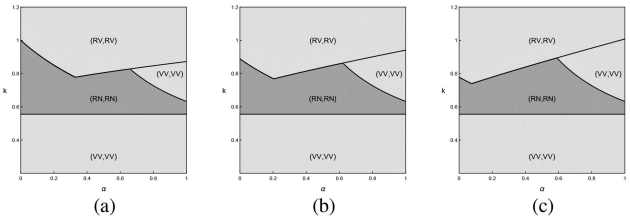


Fig. 6. Equilibria in duopoly with spillover effect ($b = 0.8$). (a) $\delta = 0.1$. (b) $\delta = 0.2$. (c) $\delta = 0.3$.

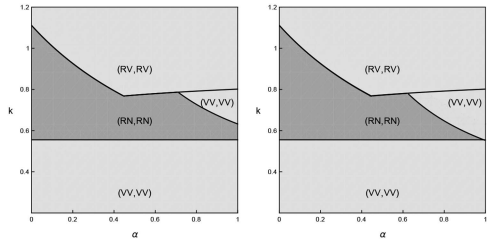


Fig. 7. Equilibria comparison in duopoly case with nonzero virtual livestreaming cost ($b = 0.8$). (a) $c = 0$. (b) $c = 0.003$.

A. *Spillover Effect*

In this section, we consider that a seller adopting the real-person livestreaming can create a spillover effect, thereby benefiting the virtual livestreaming. In this case, the real-person's effort not only affects the peak-hour demand, but also affects the demand during off-peak hours.

Note that such an effect can reflect two facts. First, consumers may watch real-person livestreaming but purchase during off-hours. According to our industry collaborator, they do observe a small portion of consumers behave in such a way, i.e., watching the real-person livestreaming but buying through the virtual livestreaming channel. Second, the real-person livestreaming quality can also increase the virtual livestreaming quality. In practice, AI agents' character and livestreaming content are typically generated based on human characters and their corresponding livestreaming content. Therefore, a high quality of real-person livestreaming can also increase the consumers' willingness to pay while observing their AI counterpart, thus boosting off-peak-hour demand.

To capture the spillover effect, we make the following changes to the inverse demand functions for the RV strategy:

$$p = 1 + e - q_g^R \tag{13}$$

$$p = 1 + \delta e - q_o^V \tag{14}$$

where $\delta > 0$ denotes the degree of spillover effect. It is clear that the real-person livestreaming effort e can increase the demand during off-hours by δe.

We completely analyze the optimal decisions and investigate the impact of the spillover effect. For the ease of exposition, we relegate all the analyses to Appendix C1 of the Supplementary Material. We use the equilibrium under intense competition ($b = 0.8$) as an example to illustrate; see Fig. 6. First, Fig. 6 shows that all our main results remain in this case. Specifically, when the degree of spillover effect increases, the platform allows the virtual livestreaming when k is either high or low, but bans it when k is medium. Moreover, under intense competition ($b = 0.8$), the unexpected equilibrium (VV, VV) still occurs when k is medium and α is large. Second, Fig. 6(a)–(c) reveals that when the degree of spillover effect increases, the (RV, RV) strategy region expands even for small α. This is because as the degree of spillover effect increases, the optimal price, effort quantity, and sellers' and platform's profits of the (RV,RV) strategy increase, i.e., $\frac{\partial p^*}{\partial \delta}, \frac{\partial e^*}{\partial \delta}, \frac{\partial q^*}{\partial \delta}, \frac{\partial \pi_s^*}{\partial \delta}, \frac{\partial \pi_p^*}{\partial \delta} > 0$. This indicates that because of the spillover effect, even during off-peak hours with low traffic, implementing the virtual livestreaming is more profitable than only adopting the real-person livestreaming during peak hours for both the platform and sellers.

B. *Nonzero Virtual Livestreaming Cost*

In the base model, we assume that the platform offers free virtual livestreaming services for sellers. In this section, we relax this assumption where sellers incur a fixed subscription cost, denoted as c. In practice, JD.com and VIP.com generally offer virtual live streaming services for free, while sellers on Taobao can pay a fixed annual fee to adopt the virtual livestreaming.

First, we derive the optimal decisions in this case and show that our main results still hold when c is within a certain range. Both the equilibrium and welfare largely remain within regions of k and α under both the monopoly and duopoly settings. Second, we investigate the impact of c on the equilibrium to illustrate the change. For example, see Fig. 7, which compares the duopoly equilibrium when competition is intense ($b = 0.8$) under zero virtual live streaming cost ($c = 0$) and a nonzero cost ($c = 0.003$). As one can see, compared to the case with no fixed cost, the seller's (VV, VV) equilibrium strategy region expands when a fixed cost is introduced. This suggests that the platform is more willing to allow the virtual livestreaming for charging the subscription fee. We refer readers to Appendix C2 of the Supplementary Material for the complete analysis of this case.

C. *Quantity Competition*

In this section, we consider quantity competition among sellers rather than price competition, where sellers decide on q and e instead of p and e. To align with the practice, we assume that sellers decide production quantities for the peak and off-peak hours simultaneously and set the uniform price for both periods. Then, we can derive the linear inverse demand function. For instance, the inverse demand function for seller i under the (RV, RV) strategy is as follows:

$$p_i = 1 + \frac{e_i}{1+\alpha} - \frac{q_i + bq_{3-i}}{1+\alpha}. \tag{15}$$

Fig. 8 gives an example of the equilibria under quantity competition when $b = 0.1, 0.4, 0.8$, respectively. As one can see, the equilibrium pattern under quantity competition remains qualitatively the same as the counterpart under price competition. That is, the platform allows the virtual livestreaming when k

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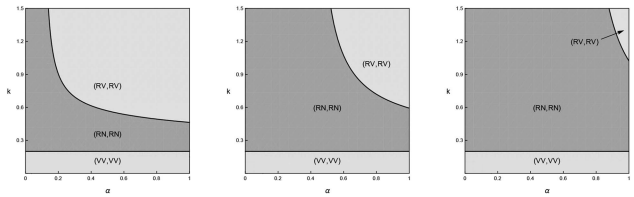


Fig. 8. Equilibria of quantity competition. (a) $b = 0.1$. (b) $b = 0.4$. (c) $b = 0.8$.

is relatively high or low, but bans it when k is medium. However, when the competition is intense ($b = 0.8$), the (VV,VV) strategy cannot become an equilibrium in the medium marginal effort cost k, different from that with price competition, as shown in Fig. 4(c). The reason is as follows. Recall that in price competition, effort first decreases then increases as competition intensifies. By contrast, as competition intensifies, sellers tend to reduce their quantity to prevent a price collapse under quantity competition. This reduction essentially lowers the marginal returns of effort, leading to optimal effort continuously decreasing. Therefore, sellers never exert excessively high effort costs, and adopting the virtual livestreaming is not more beneficial to them than the real-person livestreaming. For the same reason, in the subgame, the seller also does not fall into a Prisoner's Dilemma. We also refer readers to Appendix C3 of the Supplementary Material for the complete analysis under quantity competition.

VII. Conclusion

The virtual livestreaming automatically broadcast by AI agents has become a new and popular livestreaming service. While many E-commerce platforms commonly advocate for virtual livestreaming, some platforms hold opposite attitudes by completely banning the service. In this article, we analytically investigate when the platform should ban or offer the virtual livestreaming service by considering its operational flexibility and cost efficacy. We completely characterize the equilibrium decisions for the seller(s) and the platform in both monopoly and duopoly settings and examine the welfare impactions of different regulation policies. We then generalize our results to multiple settings, including cases with the spillover effect, nonzero adoption cost, and quantity competition.

Our main findings are as follows. First, we show that offering virtual livestreaming can induce sellers to strategically reduce real-person livestreaming effort, highlighting the impact of AIGC technology on classical livestreaming operations. Second, the platform should only ban the virtual livestreaming when the real-person marginal effort cost falls between two thresholds (depending on off-peak market size). When virtual livestreaming is allowed, sellers fully rely on AI agents to livestream if the marginal effort cost is relatively low; otherwise, they combine the virtual and real-person livestreaming across off-peak and peak hours, respectively. Third, we show that allowing virtual livestreaming can always yield a triple-win for consumers, sellers, and the platform, whereas banning it may help or hurt sellers and consumers depending on the marginal effort cost and off-peak market size. Finally, in the presence of intense competition, we find that banning virtual livestreaming only harms sellers, and a new triple-win outcome where sellers fully rely on AI agents can arise.

Our findings provide useful and important managerial guidance for platforms and sellers to manage AIGC technology in livestreaming. First, since the optimal regulation policy critically depends on product characteristics reflected by the marginal effort cost, platforms should calibrate their regulation policy on AIGC technology according to product categories, avoiding a platformwise permission or prohibition policy. The platform can safely advocate for or completely rely on AI agent for livestreaming for standardized or professional products. By contrast, for those products that require substantial consumer interaction and explanation (e.g., fashion accessories or apparel, beauty and cosmetics, and personal care), the platform should restrict the virtual livestreaming to maintain the livestreaming content quality. In this sense, our results offer a potential explanation for the divergent attitudes on AIGC technology in livestreaming from the perspective of product characteristics on different platforms. Second, platforms should also tailor their regulation policies according to the degree of competition. In particular, as competition intensifies, platforms should increasingly promote virtual livestreaming as a cost-efficiency tool to prevent destructive price wars and preserve seller margins. Third, sellers should also calibrate their livestreaming operations based on both product characteristics and competitive intensity. For standardized products, sellers can delegate all livestreaming to AI agents; while for medium- and high-customized products, it is important to reserve real-person livestreaming effort, particularly in peak hours. Under fierce competition, sellers should proactively adopt virtual livestreaming to avoid being trapped in costly effort escalation against rivals. Overall, our results underscore the strategic value and impacts of AIGC technology in livestreaming services.

Acknowledgment

The authors would like to thank the Editor-in-Chief, Jason Choi, Department Editor, and three anonymous reviewers for their constructive suggestions and comments, which have largely improved the quality of this article.

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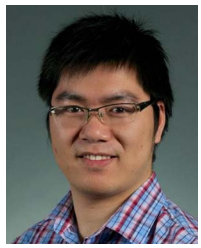
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